Cyber/Information Security Research

Autonomous Cyber Defense (ACD) – Threat-hunting toolkit

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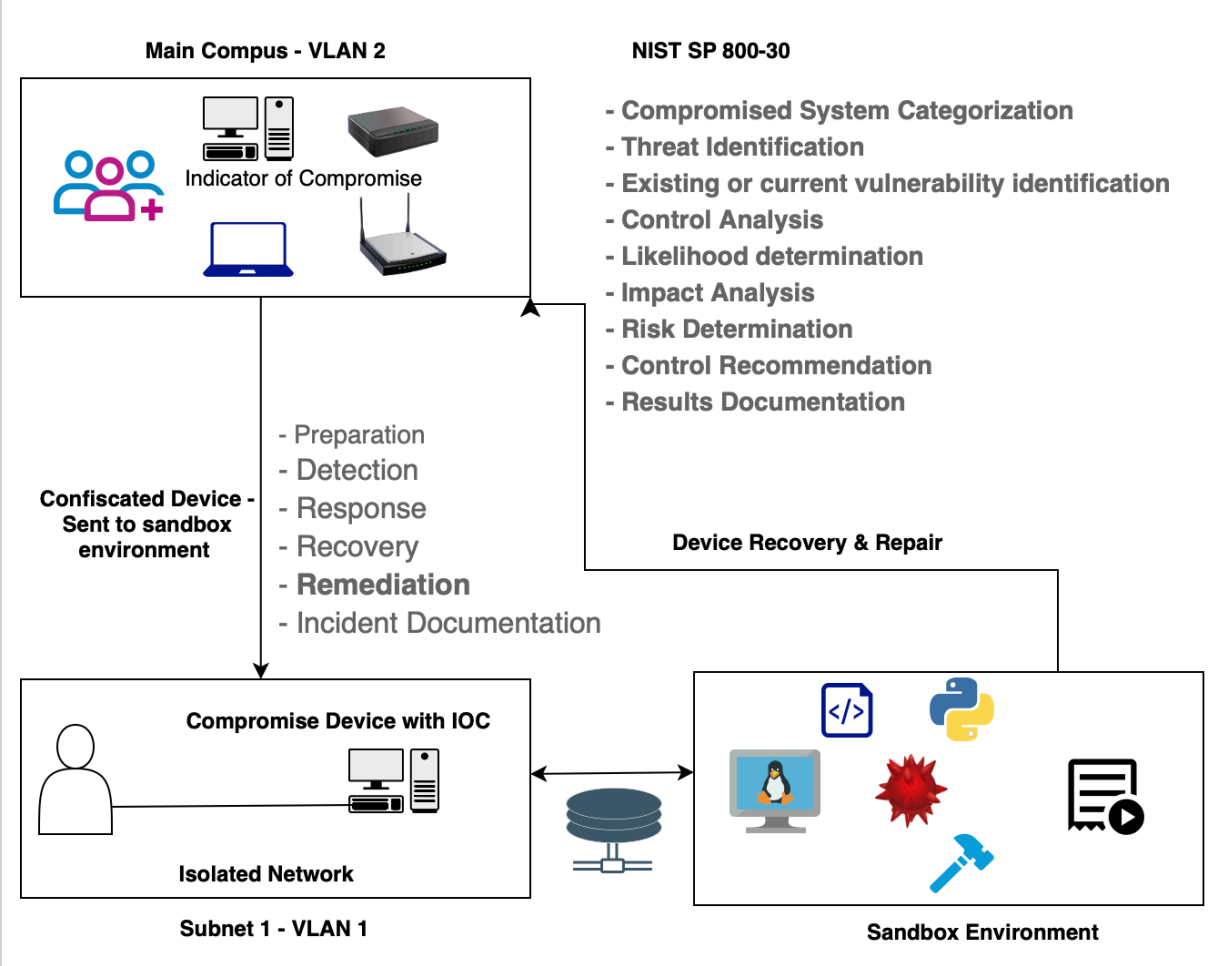
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Threat-hunt Toolkit

Objective

With the current need for autonomous cyber defense techniques and methodology utilized to reduce risk and eradicate as much threat as possible, in a realistic approach – The main objective and use-case for a post-incident response activity & threat hunting with automated response and proactive vulnerability discovery is to cut down the time it takes to investigate and analyze a threat on a network, as well as to reduce analyst burnout with manual and ad-hoc effort at tackling cyber security events.

Figure 1.0: Threat-hunt automated response approach



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Threat-hunt Toolkit

Framework and enterprise solution utilized to put to effect, the threat hunt toolkit, are listed below as a direct need to autonomous cyber defense for security threat event resolution and activity:

1. *NIST SP 800 30*
2. *NIST CSF*
3. *Incident Management Lifecycle*
4. *System Development Lifecyle (SDLC)*
5. *PLAN-DO-STUDY-ACT (PDSA)*

Overview

Practical lab environment with scenarios pertaining to real-time security event of an indicator of compromise affecting a host in a virtual environment. Additional scenario pertaining to a real-life security event affecting a host will also be illustrated within an on-premises infrastructure environment to give a realistic idea of how autonomous cyber defense and automated incident response approach is conducted in any organization. Below is the list of technologies utilized and the role it plays in autonomous cyber-defense methodology for active threat hunting.

Table 1.0: Technology Utilized

Technology	Area Utilized	Total Count
MySQL	Infrastructure/Security/Sandbox/Database	1
Python File	Infrastructure/Security/Sandbox/	3
Oracle Virtual Box	Infrastructure/Security/Sandbox/	1
Azure Cloud	Infrastructure/Security/Sandbox/Storage/Database	1
Azure SQL Database	Infrastructure/Security/Sandbox/Database	1

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Virtual Machines (on-prem)	Infrastructure/Security/Sandbox/	2
Virtual Machines AZ	Infrastructure/Security/Sandbox/Storage/Database	4
Redis	Infrastructure/Security/Sandbox/Database	2
Ansible	Infrastructure/Security	2
ClamAV	Infrastructure/Security/Sandbox/	2
VS-Code	Coding Environment	1
Azure VPN (PTS)	Infrastructure/Security/Sandbox/Storage/Database	1
Snort (IDS)	Infrastructure/Security/Sandbox	1

Purpose

During post incident response activity, it is crucial to thoroughly and carefully understand an intrusion and its activity based on the related MITRE ATT&CK framework utilized for attack methodology and lateral movement - As well as how it propagates in a sandbox environment. A compromised host isolated from the main campus VLAN will be added to a group to be allowed on the isolated network for post incident response activities. This activity usually comes next after careful remediation and containment of an indicator of compromise on a compromised host (zombie machine). Also, a quick threat hunt within the campus VLAN will also be substantial to ensure no risk residual or exposure of other important asset to the indicator of compromise (IOC). Overall, this process will be illustrated in the upcoming section below.

Threat-Hunt

Threat hunting is typically expected during and after a security event but varies based on multiple circumstances. For an incident related to a Denial-of-Service attack according to the MITRE ATT&CK framework should most likely be remediated prior to post incident response

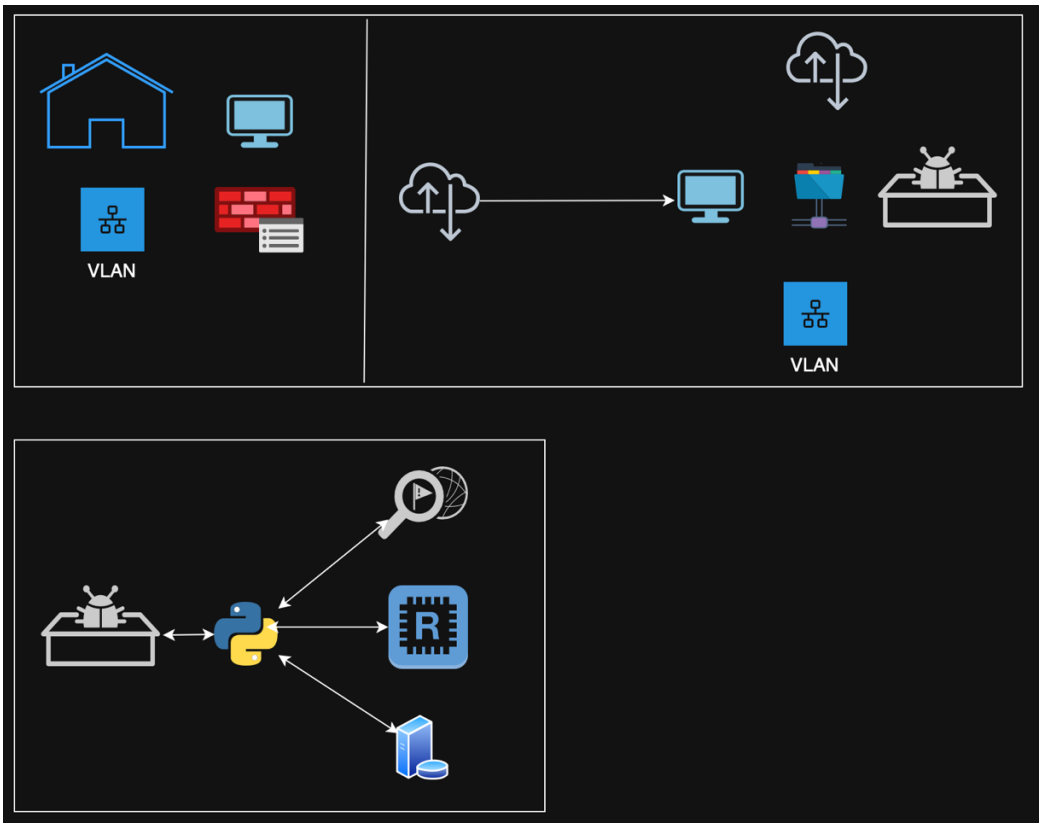
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activity. For this particular scenario, a Threat-Hunt toolkit developed in Python and utilizing data ingested from an IDS log and a master spreadsheet (comprises of data from firewall, email defense solution and more) will be put to use to examine and reverse engineer a malware behavior affecting a host in an isolated environment with the IOC thoroughly examined and detonated on a sandbox within the same isolated network.

Threat-hunt toolkit (Development/Integration)

Toolkit utilized for this use-case was designed, implemented, integrated, and tested for the purpose of examining and reverse engineering an indicator of compromise on a compromised host within the sandbox environment on a standalone network with a private internet access from an internet service provider (ISP).

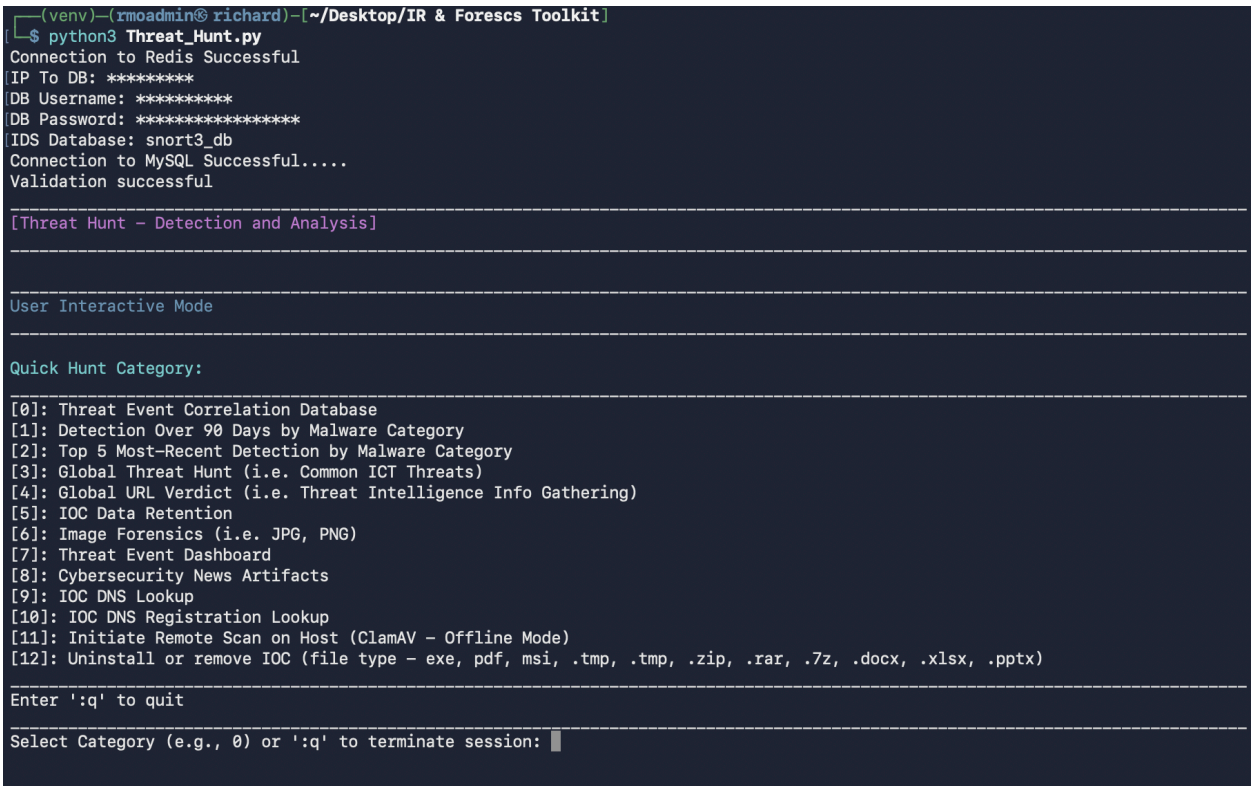
Figure 2.0: Toolkit utilization and use-case



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Threat-hunt Toolkit

Figure 3.0: Threat-hunt Toolkit



API Utilized:

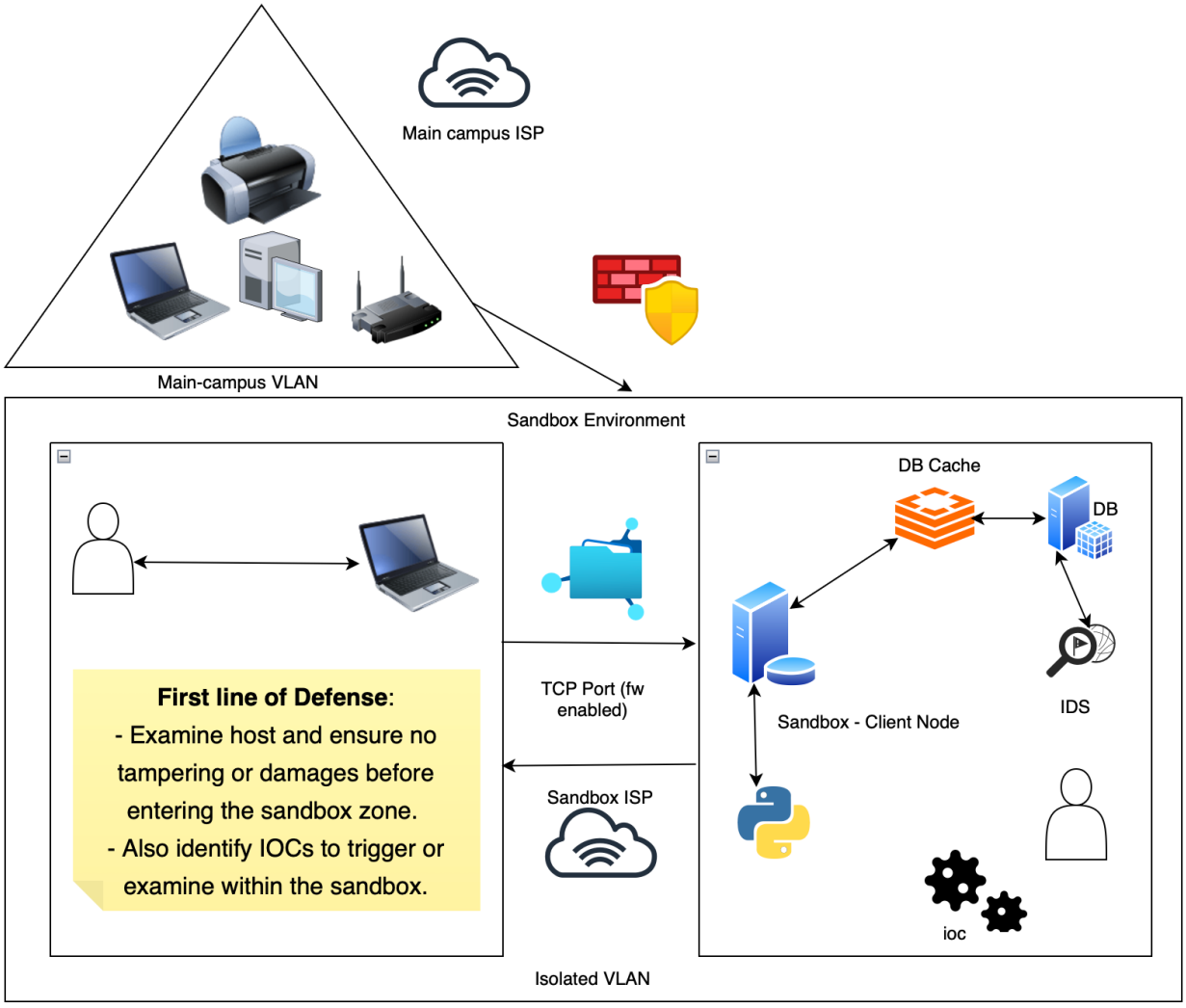
- Cloudmersive – Option 3, 4, 6
- The New York Times – Option 8

Dataset:

- IDS Database
- Master Spreadsheet – Consist of all reports generated from security solution utilized.

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Figure 4.0: Sandbox environment



In most scenario, and, during a thorough incident response post-activity, a good engineering approach is to implement and establish a secure means of interacting with an indicator of compromise (IOCs) in a sandbox environment, as well as a good approach at secure transfer of an IOC into a sandbox environment - To be reviewed and detonated on a sandbox for the purpose of malware reverse engineering - While maintaining and carefully isolating compromised devices from the rest of the asset on the network (see figure 1.0 for more detail). For the scenario

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illustrated in this research paper, two nodes are acquired, onboarded and utilized to establish a network communication on necessary TCP/IP ports required for client and server communication with the threat-hunt Python script and the database server to the IDS running within the sandbox environment. Both nodes are isolated from the main-campus network, and only accessible on the isolated network with a standalone network access from an internet service provider (ISP). In addition to the script, for faster communication and reduction of latency, Redis server is also utilized to temporarily cache the database data in the sandbox for faster communication with the python script on the client host.

Scenarios

Endpoint malware activity **-** User downloaded a text file from a rebate website (violated acceptable use policy of the organization).

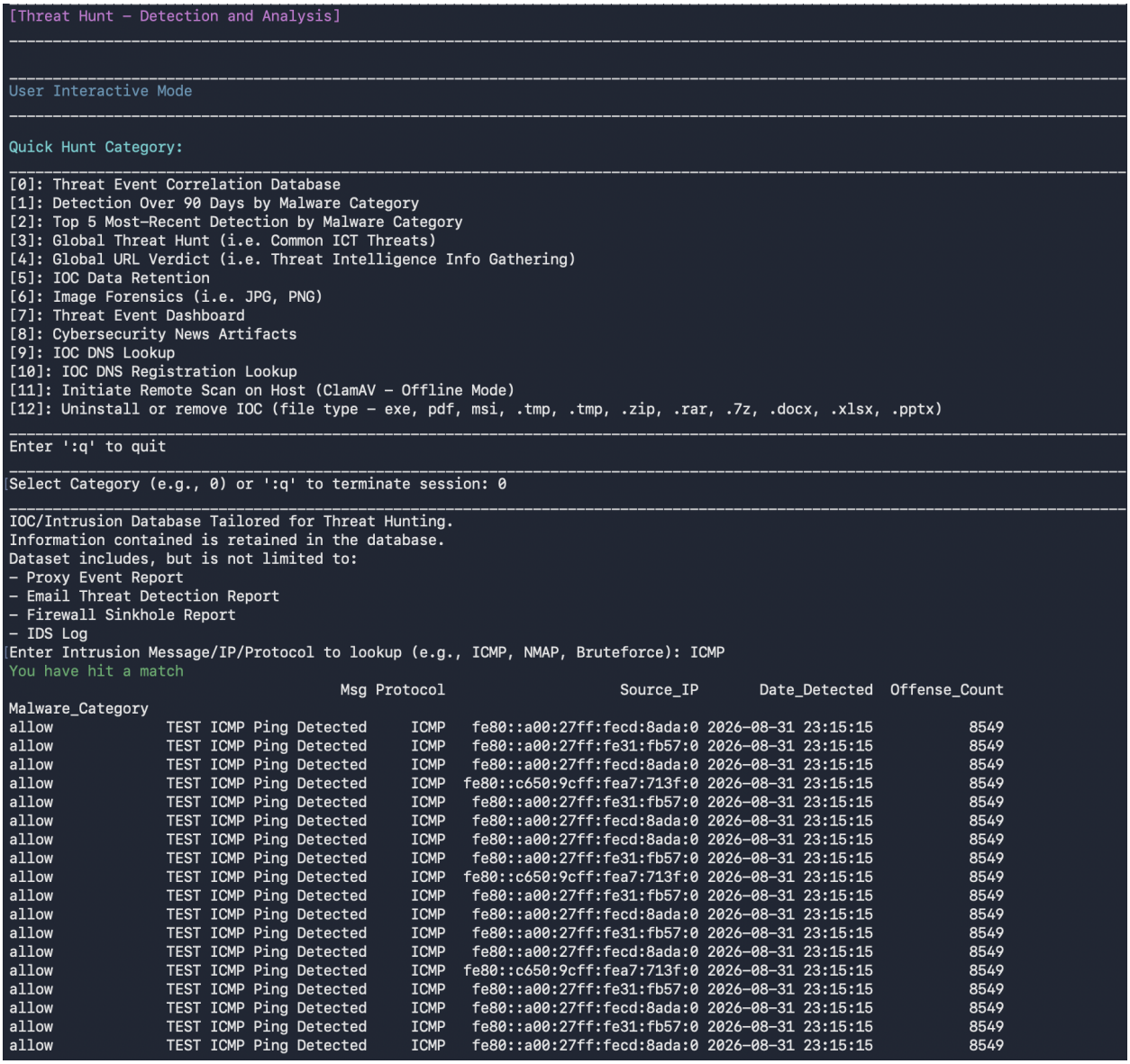
Description: Compromised device was immediately isolated from the main campus network and network arrangement was made to transport the device to the isolated VLAN. Analyst looked through the device, interviewed the end-user, and was able to gather some necessary information regarding a file that was downloaded on a rebate website. Device was contained and reimaged and IOC was transported to the sandbox environment.

Findings: IOC file had a script that attempted a ping-of-death to the next node on the same subnet as the zombie machine. The next node on the isolated network happens to be a database server for an IDS sensor (where all the IDS traffic is created and stored securely).

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Threat-hunt Toolkit

Figure 5.0: IDS Event (Ping of Death)



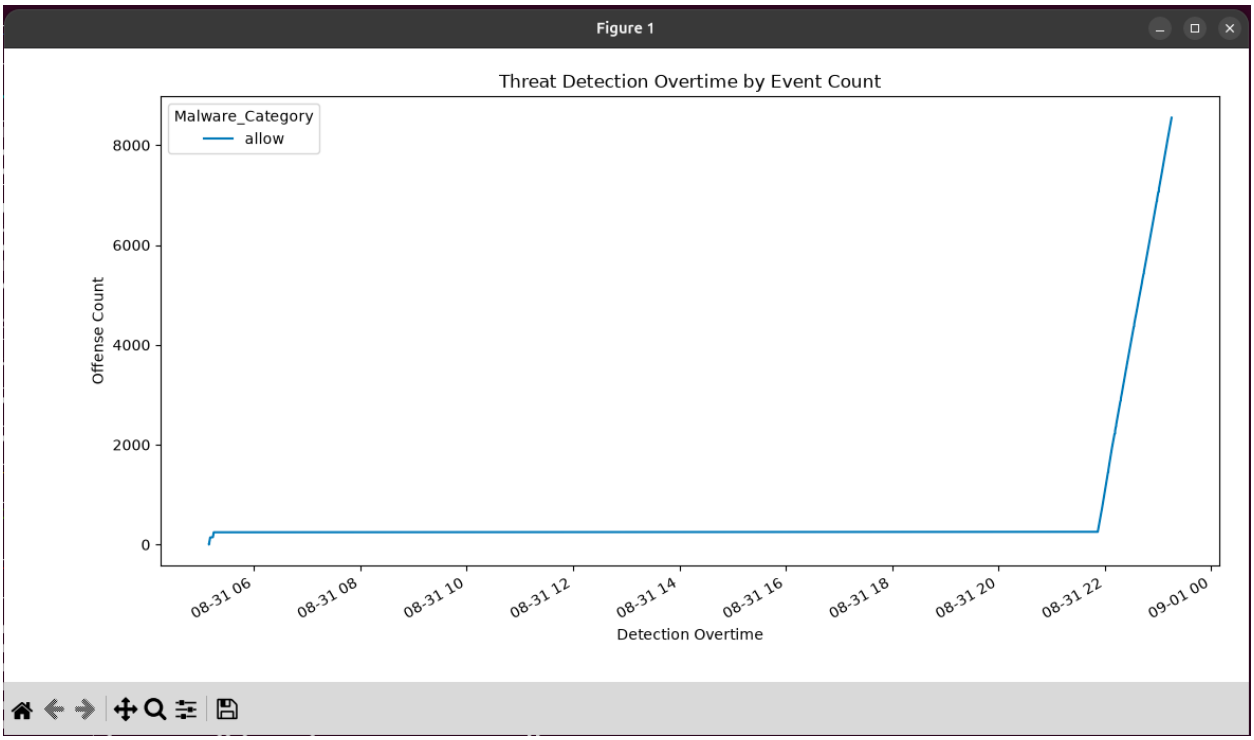
```
[Threat Hunt - Detection and Analysis]
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------------------------------------------------------------------------------------------------------------------------------------------------------
User Interactive Mode
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Quick Hunt Category:
------------------------------------------------------------------------------------------------------------------------------------------------------
[0]: Threat Event Correlation Database
[1]: Detection Over 90 Days by Malware Category
[2]: Top 5 Most-Recent Detection by Malware Category
[3]: Global Threat Hunt (i.e. Common ICT Threats)
[4]: Global URL Verdict (i.e. Threat Intelligence Info Gathering)
[5]: IOC Data Retention
[6]: Image Forensics (i.e. JPG, PNG)
[7]: Threat Event Dashboard
[8]: Cybersecurity News Artifacts
[9]: IOC DNS Lookup
[10]: IOC DNS Registration Lookup
[11]: Initiate Remote Scan on Host (ClamAV - Offline Mode)
[12]: Uninstall or remove IOC (file type - exe, pdf, msi, .tmp, .tmp, .zip, .rar, .7z, .docx, .xlsx, .pptx)
------------------------------------------------------------------------------------------------------------------------------------------------------
Enter ':q' to quit
------------------------------------------------------------------------------------------------------------------------------------------------------
Select Category (e.g., 0) or ':q' to terminate session: 0
------------------------------------------------------------------------------------------------------------------------------------------------------
IOC/Intrusion Database Tailored for Threat Hunting.
Information contained is retained in the database.
Dataset includes, but is not limited to:
- Proxy Event Report
- Email Threat Detection Report
- Firewall Sinkhole Report
- IDS Log
Enter Intrusion Message/IP/Protocol to lookup (e.g., ICMP, NMAP, Bruteforce): ICMP
You have hit a match
                                Msg Protocol                    Source_IP       Date_Detected  Offense_Count
Malware_Category
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fe31:fb57:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP  fe80::c650:9cff:fea7:713f:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fe31:fb57:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fe31:fb57:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP  fe80::c650:9cff:fea7:713f:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fe31:fb57:0 2026-08-31 23:15:15           8549
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allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fe31:fb57:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP  fe80::c650:9cff:fea7:713f:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fe31:fb57:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fe31:fb57:0 2026-08-31 23:15:15           8549
allow             TEST ICMP Ping Detected     ICMP   fe80::a00:27ff:fecd:8ada:0 2026-08-31 23:15:15           8549
```

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Threat-hunt Toolkit




```
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[10]: IOC DNS Registration Lookup
[11]: Initiate Remote Scan on Host (ClamAV - Offline Mode)
[12]: Uninstall or remove IOC (file type - exe, pdf, msi, .tmp, .tmp, .zip, .rar, .7z, .docx, .xlsx, .pptx)

Enter ':q' to quit

Select Category (e.g., 0) or ':q' to terminate session: 2

Enter Malware or Verdict Category to filter: allow
                                Msg                   Source_IP Protocol       Date_Detected    id
Malware_Category
allow             TEST ICMP Ping Detected   fe80::a00:27ff:fecd:8ada:0     ICMP 2026-08-31 23:15:15  8549
allow             TEST ICMP Ping Detected   fe80::a00:27ff:fe31:fb57:0     ICMP 2026-08-31 23:15:15  8548
allow             TEST ICMP Ping Detected   fe80::a00:27ff:fecd:8ada:0     ICMP 2026-08-31 23:15:15  8547
allow             TEST ICMP Ping Detected   fe80::a00:27ff:fe31:fb57:0     ICMP 2026-08-31 23:15:15  8546
allow             TEST ICMP Ping Detected  fe80::c650:9cff:fea7:713f:0     ICMP 2026-08-31 23:15:15  8545
```

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Further Action Taken:

File was removed, sandbox was scanned and examined for any residual to add to the post incident response activity report.

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[Threat Hunt - Detection and Analysis]
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User Interactive Mode
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[12]: Uninstall or remove IOC (file type - exe, pdf, msi, .tmp, .tmp, .zip, .rar, .7z, .docx, .xlsx, .pptx)
------------------------------------------------------------------------------------------------------------------------------------------------------
Enter ':q' to quit
------------------------------------------------------------------------------------------------------------------------------------------------------
Select Category (e.g., 0) or ':q' to terminate session: 11
------------------------------------------------------------------------------------------------------------------------------------------------------
Enter hostname or IP address to scan: 10.0.0.72
/tmp/20260903_223903-scantemp.64e5198840/html-tmp.ba6040e9f6/notags.html: OK
/tmp/20260903_223903-scantemp.64e5198840/html-tmp.ba6040e9f6/nocomment.html: OK
/tmp/promocode.txt: OK
/tmp/.X0-lock: OK

----------- SCAN SUMMARY -----------
Known viruses: 3628028
Engine version: 1.4.6
Scanned directories: 13
Scanned files: 4
Infected files: 0
Data scanned: 0.24 MB
Data read: 0.10 MB (ratio 2.48:1)
Time: 4.433 sec (0 m 4 s)
Start Date: 2026:09:03 22:43:43
End Date:   2026:09:03 22:43:48


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```

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```
Quick Hunt Category:
_____________________________________________________________________________________________________________________
[0]: Threat Event Correlation Database
[1]: Detection Over 90 Days by Malware Category
[2]: Top 5 Most-Recent Detection by Malware Category
[3]: Global Threat Hunt (i.e. Common ICT Threats)
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[12]: Uninstall or remove IOC (file type - exe, pdf, msi, .tmp, .tmp, .zip, .rar, .7z, .docx, .xlsx, .pptx)
_____________________________________________________________________________________________________________________
Enter ':q' to quit
_____________________________________________________________________________________________________________________
Select Category (e.g., 0) or ':q' to terminate session: 12
_____________________________________________________________________________________________________________________
Enter the full path of the IOC file to uninstall/remove: /tmp/promocode.txt
Successfully removed: /tmp/promocode.txt
```

Risk Quantification – Single Loss Expectancy

Compromised asset was classified as high to critical due to other assets connected on the same subnet - As well as the leg of the **CIA** triad being disrupted (***Availability***). It is important to understand the percentage of asset(s) lost should the indicator of compromise had successfully initiated a ping-of-death on assets connected on same subnet as the zombie machine.

Table 2.0: Single Loss Expectancy

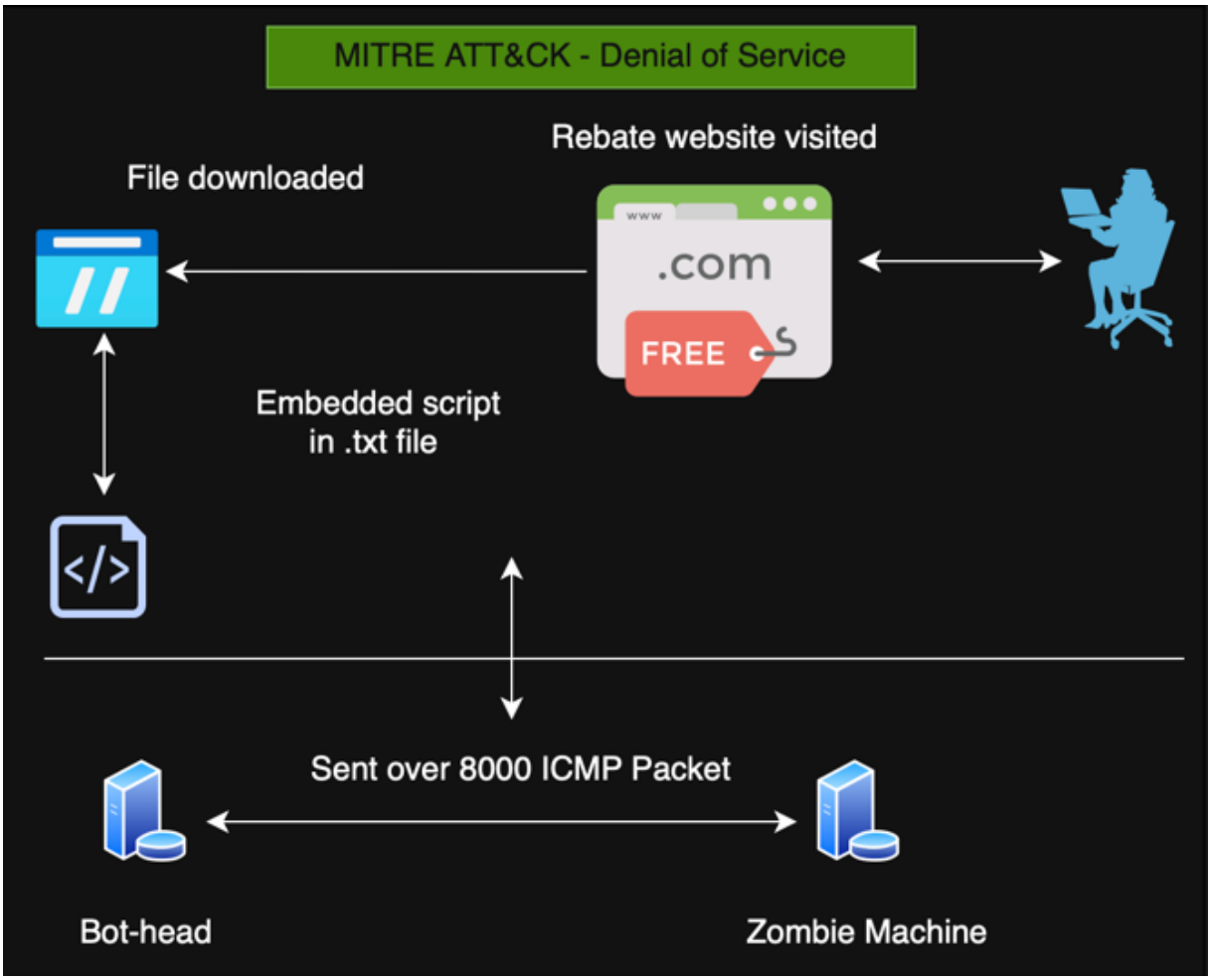
Asset	Asset Value	Exposure Factor	Single Loss Expectancy
Printer (X-40)	20,000 USD	100%	20,000 USD
Servers (X-20)	40,000 USD	100%	40,000 USD
Workstations (X-40)	30,000 USD	100%	30,000 USD
Other IOT (X-100)	100,000 USD	100%	100,000 USD
-	-	-	Total: $190,000

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MITRE ATT&CK Tactics

Attack vector(s) and threat indicator(s) utilized during a security event is crucial to understand, to achieve positive and proactive result of an overall malware reverse engineering approach.

- **Related Attack Tactics**: Denial of Service
- **Impact:** Availability – VM Went into Crash-mode on target host
- **Severity**: High-to-Critical



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Threat-hunt Toolkit

References

- https://github.com/rmo616/Technical-Professional-Documents/blob/main/Technical%20Portfolio_RMO.pdf
- https://github.com/rmo616/IR_Forensics_Toolkits (Available upon request)
- https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-30r1.pdF
- https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r3.pdf

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